

Hannah Arnold

LinkedIn: <https://www.linkedin.com/in/hannah-arnold-848b372a7/>

I'm a First-Class AI and Robotics graduate with hands on experience developing embedded software in C using STM32 microcontrollers. Through university projects I have developed skills in embedded programming, integrating sensors, motors and peripherals while debugging hardware and software issues. I enjoy solving technical problems and am looking to begin my career in graduate embedded programming or software development.

Education

Staffordshire University September 2023 - May 2026

Qualification	Grade
Bachelor of Sciences (Hons) AI and Robotics	First Class Honours Degree

Relevant Modules

- Robotic Programming
 - Developed embedded applications in C for STM32 microcontrollers
 - Programmed GPIO, timers and hardware peripherals
 - Debugged software and hardware issues during development
 - Tested embedded applications on physical hardware
- Final Year Project
 - Designed, built and tested an embedded system using STM32 Microcontrollers programmed in C
 - Integrated embedded software with hardware components including multiple sensors, push buttons and a stepper motor
 - Debugged software and hardware issues throughout development
 - Produced technical documentation and project report
- AI and Chatbots
 - Developed software solutions using Python and C++
 - Gained experience in algorithm design and structured programming

City of Stoke-on-Trent Sixth Form College 2021 – July 2023

Qualification	Grade
Maths A level	B
Maths Fast Track A level	D
Extended Project Qualification (Artificial Intelligence)	B
Physics A level	D

Technical Skills

- Languages: C and C++, Python
- Microcontrollers: STM32
- Embedded Systems: GPIO, Timers, Hardware Interfacing, Sensor Integration and Motor Control

Some Personal Details Have been removed for safety

- Device Communication: I2C
- Tools: STM32CubeIDE, Git, GitHub and Visual Studio Code

Projects

Final Year Project - Implementing microcontrollers and sensors to determine ride path on roller coasters October 2025 - May 2026

This project was developed in order to explore how user experience could be enhanced on roller coasters without the use of virtual reality. It used multiple sensors and a camera to collect data from the rider, then uses this to determine emotion and choose a suitable path for the rider.

- Developed embedded systems in C using STM32 Microcontrollers
- Interfaced multiple sensors
- Debugged hardware and firmware issues
- Designed and tested electronic circuits

Study Progression Website May 2026 - Present

This project is a record of my progress as I continue to develop my technical skills after graduating. The site will showcase projects and experiments covering the technologies and concepts I'm exploring from web development such as semantic HTML to embedded programming and other areas of software engineering.

- <https://xxhansterxx.github.io/>
- Used Git and GitHub for version control and management of software projects
- Built a personal portfolio website to document and demonstrate ongoing development
- Published projects and experiments across web development and embedded programming

Employment

Retail Assistant - Primark April 2022 - Present

- Worked in a fast-paced retail environment delivering excellent customer service
- Developed strong communication and teamwork skills
- Adapted quickly to changing priorities while maintaining accuracy
- Trusted to work across multiple departments and tills

Hobbies

University Engineering Society September 2023 – May 2024

- Member of the University Engineering Society, taking part in weekly practical sessions
- Gained hands-on experience with electronics, including soldering LEDs
- Used CAD software for 3D printing and laser cutting projects
- Developed practical engineering through workshop-based activities

Macclesfield Warriors Hockey Team Committee and Captain. December 2024 - Present

- Led a competitive team during matches and training
- Organised team communication as communication manager on the teams committee
- Developed leadership, organisation and decision-making skills
- Handled conflicts calmly and effectively in a high-pressure environment

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